

Attention and recall under time pressure

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Abstract

We tested whether time pressure changes recall. Ninety-six people took part. Recall fell under pressure, and the effect held across two measures. We report every test in full, so that the analysis can be checked by a reader.

Method

Ninety-six adults took part, and each of them completed both conditions. We measured recall as the number of items reported, and we measured confidence on a seven-point scale. The study was not preregistered.

Results

Reaction times differed between the groups, $t(23) = 2.45$, $p = .022$.
The effect of condition on recall was reliable, $F(2, 30) = 5.10$, $p = .012$.
The effect did not depend on the order of the blocks.

Accuracy differed between the conditions, $\chi^2(1, N = 223) = 8.69$, $p = .003$.
Confidence rose with accuracy, $r = .42$, $p = .02$, although the two measures were collected on different days.

Recall under pressure was lower than recall at rest, $t(46) = 1.80$, $p = .04$.
The same pattern appeared in the second sample, $F(1, 118) = 9.20$, $p = .003$.

The manipulation check succeeded, $z = 1.96$, $p = .05$. Participants who noticed the manipulation reported more effort, $t(19) = 4.15$, $p < .001$.

Self-reported tiredness rose over the session, $F(3, 92) = 2.71$,
 $p = .049$, although the measure was collected only twice.

Table 1. Recall by condition

Condition	N	M	SD	t	p
Rest	48	14.2	3.1	2.45	.022
Pressure	48	11.8	3.6	1.80	.040

Discussion

Time pressure lowered recall in both samples. The effect is small and the design is not conclusive. We report it so that others can test it in a larger sample.

References

- Example, A., & Sample, B. (2019). Attention under load. *Journal of Invented Findings*, 12(3), 145-162.
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- Sample, B., Example, A., & Fictional, C. (2020). Measuring effort.

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